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SUPERSEDING
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MILITARY SPECIFICATION

CONNECTOR, ELECTRICAL, BOOSTER-TO-MISSILE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of electrical connector for use on a guided missile.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization/Documentation Division (524), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1420

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PPP-B-636	Box, Shipping, Fiberboard
PPP-B-665	Boxes, Paperboard, Metal Edged and Components
PPP-B-676	Boxes, Set-Up
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation Packing, Methods of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper Overlaid
MIL-L-10547	Liners, Case, and Sheet, Overwrap, Water-Vaporproof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWING

Naval Sea Systems Command

(Code Ident 10001) 2178133	Connector, Electrical, Booster to Missile

PUBLICATION

Naval Sea Systems Command

(Code Indent 10001) CD 2178133	Classification of Defects for Connector, Electrical, Booster to Missile

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. The issue of the document which is indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. Unless otherwise specified (see 6.2.1), a first article shall pass the inspections specified in 4.3.

3.2 Construction. The connector shall be constructed in accordance with Drawing 2178133(10001). The connector shall comply with all the requirements of the drawing and shall meet the requirements specified herein.

3.3 Performance. The connector shall meet all of the performance requirements given herein.

3.3.1 Continuity and resistance. Each electrical conductor, listed in Table I, shall maintain continuity and shall have the resistance specified when a potential of 28 volts direct (Vdc) is applied to that conductor.

3.3.2 Insulation breakdown. Each conductor of the connector shall withstand 1500 Vdc for at least 0.10 second without evidence of breakdown damage or arcing.

3.3.3 Leakage resistance. After having been tested for the insulation breakdown requirement of 3.3.2, the open circuit leakage resistance between all conductors and between each conductor and ground shall be not less than 20 megohms, when a potential of 500 Vdc is applied.

3.4 Identification. Each connector shall be identified in accordance with MIL-STD-130 and Drawing 2178133(10001).

3.5 Workmanship. The workmanship and finish of the connector shall be of sufficiently high grade to ensure satisfactory operation consistent with the requirements of this specification. The connector shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign matter.

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TABLE I. Continuity and resistance.

Conductors			Resistance (ohm max.)
J1-1	to	J2-A	0.700
J1-2	to	J2-B	0.700
J1-3	to	J2-C	0.700
J1-4	to	J2-D	0.125
J1-5	to	J2-E	0.700
J1-6	to	J2-F	0.700
J1-7	to	J2-H	0.700
J1-8	to	J2-J	0.700
J1-9	to	J2-K	0.125
J1-10	to	J2-L	0.700
J1-11	to	J2-M	0.700
J1-12	to	J2-N	0.700
J1-13	to	J2-P	0.700
J1-14	to	J2-R	0.700
J1-15	to	J2-S	0.700
J1-16	to	J2-T	0.700
J1-17	to	J2-U	0.700
J1-18	to	J2-V	0.700
J1-19	to	J2-W	0.700
J1-20	to	J2-X	0.700
J1-21	to	J2-Y	0.700
J1-22	to	J2-Z	0.700
J1-23	to	J2-a	0.700
J1-24	to	J2-E1	0.700

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order (see 6.2.1), the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

c. Inspection of packaging (see 4.5)

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4.3 First article. Unless otherwise specified (see 6.2.1), one unit selected at random from those first manufactured under each contract, shall be designated as a first article. The first article shall be manufactured using the same manufacturing conditions and processes as are proposed for all units furnished under the same contract or purchase order.

4.3.1 First article inspection. The first article of 4.3 shall be subjected to an inspection, which shall consist of the quality conformance inspections, with the following exception and addition:

a. Sampling plans shall not apply. All first article units shall be subjected to all specified inspections.

b. In addition, all first article assemblies shall be inspected for compliance with all characteristics of Drawing 2178133.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of inspections as specified in Table II, conducted in any order. When specified in the contract (see 6.2.1), the contractor shall prepare a report giving the results obtained for all inspections and tests performed.

4.4.1 Quality conformance inspection lot. A quality conformance inspection lot shall consist of all units which are manufactured under essentially the same conditions and are offered for inspection at any one time.

4.4.2 Quality conformance sampling. No sampling shall be allowed for quality conformance inspection. All units shall be subjected to all specified inspections in Table II.

4.4.3 Action in case of quality conformance failures. Inability of a unit to meet any requirement of the quality conformance inspections shall constitute a failure. Defective units found during quality conformance inspection shall be rejected.

TABLE II. Quality conformance inspections.

Examination or test	Requirement paragraph	paragraph
Visual and mechanical examination	3.2	4.7.1
Continuity and resistance	3.3.1	4.7.2.1
Insulation resistance	3.3.2	4.7.2.2
Leakage resistance	3.3.3	4.7.2.3

4.5 Inspection of packaging. Sample units which have been packaged or packed shall be selected and inspected in accordance with MIL-P-116, to verify compliance with the requirements of Section 5 of this specification.

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4.6 Inspection equipment and facilities. Inspection equipment and facilities shall be of adequate quality and quantity to permit performance of the required inspections. Unless otherwise specified, inspection equipment error shall be not greater than 10 percent of the tolerance on any measurement for which the equipment is used. Where SMI are used, they shall be accurate within the following limits.

a. Instruments used to measure dimensions which are expressed to 3 decimal places --- 0.0001 inch.

b. Instruments used to measure dimensions which are expressed to less than 3 decimal places --- 0.001 inch.

4.6.1 Inspection conditions. Unless otherwise specified, all inspections shall be performed under the following conditions of normal pressure, temperature and humidity:

a.	Barometric pressure	28 to 32 inches of mercury
b.	Temperature	16 to 35 C (60 to 95 F)
c.	Relative humidity	10 to 95 percent

4.7 Inspection Methods.

4.7.1 Visual and mechanical examination. The inspector shall verify that the materials used to construct the unit are in accordance with 3.2, that the marking is in accordance with 3.4, and that the unit meets the workmanship requirements of 3.5.

4.7.1.1 Inspection of attributes. All physical attributes, both dimensional and non-dimensional listed on CD, 2178133 shall be inspected as specified on the CD except that no sampling shall be allowed. For dimensional attributes specified on the drawing but not listed on the CD, inspections shall be performed and sampling is permitted. Sampling shall be in accordance with MIL-STD-105 on plans approved by the Government. The approved AQL's shall apply to the individual attributes and not the total of all attributes.

4.7.2 Performance inspection methods. The following tests shall be performed on the connector.

4.7.2.1 Continuity and resistance. Apply 28 Vdc to each of the conductors successively, listed in Table I. Using a resistance measuring device, verify the continuity of each conductor and measure the resistance of each conductor. Each conductor shall show continuity and the resistance of each conductor shall be as specified in Table I.

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4.7.2.2 Insulation breakdown. Apply 1500 Vdc for not less than 0.10 second between each conductor and all other conductors connected together and grounded. Each connector shall be examined for evidence of arcing or breakdown damage.

4.7.2.3 Leakage resistance. Following the insulation resistance test of 4.7.2.2, apply 500 Vdc minimum, between all conductors and between each conductor and ground. Using a megohmmeter device, measure the open circuit leakage resistance. The leakage resistance shall be not less than 20 megohms.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unless otherwise specified (see 6.2), connectors shall be individually protected and unit-packaged in accordance with Method IA of MIL-P-116, without the use of contact preservatives. One connector only shall be included in a unit package. Unless otherwise specified, five unit package shall be further packaged in intermediate containers conforming to PPP-B-566, Class 1 or PPP-B-636, PPP-665, or PPP-B-676. The gross weight of the intermediate container shall not exceed 20 pounds.

5.1.2 Level C. Connectors shall be preserved and packaged in accordance with the contractor's commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level C. Connectors, packaged as specified, shall be packed in overseas-type wire bound wood, wood-cleated fiberboard, wood-cleated plywood, nailed wood, fiber (Class 2), or wood-cleated paper-overlaid boxes conforming to PPP-B-585, PPP-B-591, PPP-B-601, PPP-B-621, PPP-B-636, or MIL-B-10377, respectively. Shipping containers shall have case liners conforming to MIL-L-10547: the case liners shall be closed and sealed in accordance with the appendix thereto. Case liners for boxes conforming to PPP-B-636 will not be required provided the center and edge seams and manufacturer's joints are sealed with tape, at least 1 1/2 inches wide, conforming to PPP-T-76. Box closures and strapping shall be as specified in the applicable box specification or appendix thereto. Fiber boxes conforming to PPP-B-636 may be banded with tape conforming to type IV of PPP-T-97 and appendix thereto in lieu of steel straps. The gross weight of wood boxes shall not exceed applicable box specification.

5.2.2 Level B. Connectors, packaged as specified, shall be packad in domestic-type wirebound wood, wood-cleated fiberboard, wood-cleated paper-overlaid boxes respectively. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

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5.2.3 Level C. Connectors, packaged as specified, shall be packed in containers of the type, size, and kind commonly used for the purpose, in a manner that will ensure acceptance by common carrier and safe delivery at destination. Shipping shall comply with the Uniform Freight Classification Rules, or regulations of other carriers, as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract, unit packages, intermediate packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intened use. The connector covered is intended for use on the MK 12 MOD 1 Guided Missile.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Whether the contractor is responsible for inspection (see 4.1)

c. Whether the contractor should use his own laboratory or other suitable facility (see 4.1)

d.Whether first article inspection shall be required and whether more than one connector shall be tested (see 4.3).

e. That units subjected to first article testing shall not be delivered on the contract.

f. Whether a quality conformance test report is required (see 4.4)

g. Applicable level of preservation, packaging and packing (see Section 5).

6.2.2 Contract data-requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of 27.410-6 of the DoD FAR Supplement are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph No.	Data requirement title	Applicable DID No.
4.4	Reports, test	DI-T-2072

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(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the Contracting Officer).

Preparing Activity:
Navy - OS

Project No. 1420-N257

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